ADITYA UNIVERSITY

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PART-I

S.No: B304086

Programe	:	B.Tech
Hall Ticket No:		24B11AI235
Name	:	Mallidi Varshith Reddy
Examination	:	III SEMESTER END EXAMINATIONS REGULAR
Month-Year	:	Nov-25
Branch	:	Artificial Intelligence & Machine Learning
Course Code	:	241MA009
Course Name	:	Probability & Statistics
Date of Exam	:	18.11.2025

J. Pawan.

Signature of the Controller of Examination

M. Varshith Reddy 18/11/2025

Signature of the Student with date

[illegible] 18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

UNIT - 1

1) b) Given,

x :	0	1	2	3	4	5	6	7
$p(x)$:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

Now,

i) k

Here, we know that,

$$\Sigma p(x) = 1$$

$$0+k+2k+2k+3k+k^2+2k^2+7k^2+k = 1$$

$$9k+11k^2 = 1$$

$$11k^2+9k-1 = 0$$

$$\boxed{k = 0.1}$$

iv) mean

$$\text{mean} = \frac{0(0)+k(1)+2k(2)+3(2k)+4(3k)+5(k^2)+6(2k^2)+7(7k^2+k)}{8}$$

$$= \frac{0+k+4k+6k+12k+5k^2+12k^2+49k^2+7k}{8}$$

$$= \frac{30k+66k^2}{8}$$

$$= \frac{30(0.1)+66(0.1)^2}{8}$$

$$= \frac{3+0.66}{8} = \frac{3.66}{8}$$

$$= 0.4575$$

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v) variance

$$\sigma^2 = \frac{(0^2)(0) + (1)^2(k) + (2)^2(2k) + (3)^2(2k) + 4^2(3k) + (5)^2(k^2) + (6)^2(2k^2) + (7)^2(7k^2+k)}{8}$$

$$= \frac{0 + k + 8k + 18k + 48k + 25k^2 + 72k^2 + 343k^2 + 49k^2}{8}$$

$$= \frac{27k + 48k + 489k^2}{8}$$

$$= \frac{75k + 489k^2}{8}$$

$$= \frac{75(0.1) + 489(0.01)}{8}$$

$$= \frac{7.5 + 4.89}{8} = \frac{12.39}{8} = 1.\cancel{548}.$$

ii) $p(x<6) = p(x=0) + p(x=1) + p(x=2) + p(x=3) + p(x=4) + p(x=5)$

$= \cancel{0+k} + \cancel{4k+6}k + 1\cancel{8k+5}k^2$

$= \overset{23}{\cancel{8k}} + \overset{5}{}k^2 \quad \Rightarrow 0 + k + 2k + 3k + 2k + k^2$

$= {}^{23}8(0.1)\cancel{+5(0.1)}^2 \quad \Rightarrow 8(0.1) + 5(0.1)^2$

$= \cancel{0.8} + \cancel{0.01} \quad \cancel{2.35} \quad \Rightarrow 0.8 + 0.01$

$= \cancel{0.81}. \quad \Rightarrow 0.81.$

iii) $p(0<x<5) = p(x=1) + p(x=2) + p(x=3) + p(x=4)$

$\Rightarrow k + 2k + 2k + 3k = \cancel{k} + \cancel{2k} + \cancel{8k} + \cancel{18k}$

$\Rightarrow 8k \Rightarrow 8(0.1) = \cancel{8k} = 8(\cancel{0.1}) = \cancel{0.8}$

$= 0.8 \quad \cancel{23k} = 23(\cancel{0.1}) = \cancel{2.3}$

Q.No.

1) a) Let Event of getting firstbag $= P(A) = \frac{2}{6}$
Let Event of getting Second bag $= P(B) = \frac{2}{6}$
Let Event of getting third bag $= P(C) = \frac{2}{6}$

Now.
The probability of getting one white and one red ball From A $P(\frac{wr}{A}) = \frac{2}{3}$
The probability of getting one white and one red ball From B $P(\frac{wr}{B}) = \frac{2}{5}$
The probability of getting one white and one red ball From C
$P(\frac{wr}{C}) = \frac{2}{4}$

Now, From Baye's theorem

$$P(B/wr) = \frac{P(B) \cdot P(wr/B)}{P(A) \cdot P(wr/A) + P(B) \cdot P(wr/B) + P(C) \cdot P(wr/C)}$$

$$= \frac{\frac{2}{6} \times \frac{2}{5}}{\frac{2}{6} \times \frac{2}{3} + \frac{2}{6} \times \frac{2}{5} + \frac{2}{6} \times \frac{2}{4}}$$

$\frac{0.133}{0.522}$

$$\Rightarrow \frac{\frac{4}{30}}{\frac{4}{18} + \frac{4}{30} + \frac{4}{24}} \Rightarrow \frac{\frac{4}{30}}{\frac{47}{90}}$$

$$\Rightarrow \frac{4}{\cancel{30}} \times \frac{\overset{30}{\cancel{90}}}{47}$$

$$\Rightarrow 0.254$$

Hence, the probability that the balls drawn From Second bag is 0.254

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3)a) Given, $n = 1000$

Mean $(\mu) = 78$

and standard deviation $(\sigma) = 11$.

Now,

$$z = \frac{x - \mu}{\sigma}$$

⇒ i) $x > 90$

$$z = \frac{x - \mu}{\sigma} = \frac{90 - 78}{11} = 1.09$$

Now, ⇒ $0.5 - A(z_1)$

⇒ $0.5 - A(1.09)$

⇒ $0.5 - 0.3621$.

⇒ 0.1379.

By using the z statistical table we found that The probability of students getting Marks above 90 are 0.1379.

ii) $x < 60$

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 78}{11} = -1.636.$$

Now,

⇒ $0.5 - A(z)$

⇒ $0.5 - A(1.636)$

⇒ $0.5 - 0.4484$

⇒ 0.0516.

Hence, By using the z Statistical table we found that the probabili

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of students getting marks below 60 are 0.0516.

3) b) Given,

$n = 5$

and $p = 20\% = 0.2$

Hence we know that

$p+q = 1$

$0.2 + q = 1$

$q = 1 - 0.2$

$q = 0.8$.

Now

The Binomial expression is

$\cancel{5_{C_r}\, p^r}\, q$

The general binomial expression is

$n_{C_r}\, p^r q^{n-r}$

From the data we can write that $5_{C_r} (0.2)^r (0.8)^{5-r}$.

i) none is defective.

$P(r=0)$

The Binomial expression is

$\Rightarrow 5_{C_0} (0.2)^0 (0.8)^5$

$\Rightarrow 1(1)(0.327)$

$= (0.327)$

$\Rightarrow 0.327$.

$\therefore$ The probability that none of that is defective is 0.327.

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ii) one is defective

$p(r=1)$

The Binomial expression

$\Rightarrow 5c_1 (0.2)^1 (0.8)^{5-1}$

$\Rightarrow 5 (0.2)(0.8)^4$

$= 0.4096.$

Hence, the probability of one is defective is 0.4096

iii) At most 2 defectives.

$p(r=0) + p(r=1) + p(r=2)$

$0.327 + 0.4096 + 5c_2 (0.2)^2 (0.8)^{5-3}$

$0.327 + 0.4096 + 10 \cancel{\times}\ 0.256.$

$\Rightarrow 0.9926.$

Hence, the probability that atmost 2 are defective is 0.9926.

5)a) Data types :

The Data types are defined as the classification of data. By Based on the nature of the Data The Data types are classified. There are Classified into two type of datas. Those Quantitative data type and the other type is Qualitative datatype and the Qualitative data type is also called as the

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categorical datatype.

Qualitative data type:

It was also called as the categorical data type. There are no numerical operations are performed in this Qualitative data type. For example it will classify the data based on the gender, religious and all.

Example: gender, religion.

Quantitative data type:

The Quantitative data type was also called as the logical data type. It will classify the numericals also. It will classify in an logical sequence.

Methods of data collection.

There are two methods of data collections primary data collection and the secondary data collection.

Primary data collection:

The primary data collection store the data as per the update The primary data not be stored from the long time.

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Secondary datatype:

The ~~perio~~ previous data will be analyzed in the secondary datatype.

Hence these are the two different types of data collections.

5)b) Given, $N = 2, 3, 6, 8, 11$

$N = 5$

and sample size $= 2$.

i) Mean of the population is $\frac{2+3+6+8+11}{5} \Rightarrow \frac{30}{5} = 6$.

ii) standard deviation of the population is σ^2

$$\frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$\Rightarrow \frac{16+9+0+4+25}{5}$$

$$\Rightarrow \frac{54}{5} \Rightarrow \sigma^2 = 10.8. \Rightarrow \sigma = \sqrt{10.8}$$

iii) The Mean of the Sampling distribution of Means.

Now consider, N^n

$\Rightarrow 5^2$

$\Rightarrow 25$.

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iii) All possible samples of size 2 drawn with replacement are $N^n = 5^2 = 25$

(2, 2) (2, 3) (2, 6) (2, 8) (2, 11)

(3, 2) (3, 3) (3, 6) (3, 8) (3, 11)

(6, 2) (6, 3) (6, 6) (6, 8) (6, 11)

(8, 2) (8, 3) (8, 6) (8, 8) (8, 11)

(11, 2) (11, 3) (11, 6) (11, 8) (11, 11)

Means of the samples

2	2.5	4	5	6.5
2.5	3	4.5	5.5	7
4	4.5	6	7	8.5
5	5.5	7	8	9.5
6.5	7	8.5	9.5	11

Mean of sampling distribution

$$\mu_x = \frac{2+2.5+4+5+6.5+2.5+3+4.5+5.5+7+4+4.5+6+7+8.5+5+5.5+7+8+9.5+6.5+7.8.5+9.5+11}{25}$$

$$= \frac{150}{25}$$

$$\boxed{\mu_x = 6}$$

iv) standard deviation of sample distribution

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{25}(x_i-\bar{x})^2}{N}$$

$$= \frac{(2-6)^2+(2.5-6)^2+(4-6)^2+(5-6)^2+(6.5-6)^2+(2.5-6)^2+(3-6)^2+(4.5-6)^2+(5.5-6)^2+(7-6)^2+(4-6)^2+(4.5-6)^2+(6-6)^2+(7-6)^2+(8.5-6)^2+(5-6)^2+(5.5-6)^2+(7-6)^2+(8-6)^2+(9.5-6)^2+(6.5-6)^2+(7-6)^2+(8.5-6)^2+(9.5-6)^2+(11-6)^2}{25}$$

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$$= \frac{134.775}{25}$$

$\sigma_{\bar{x}}^2 = 5.391$

Standard deviation $\sigma_{\bar{x}} = \sqrt{5.391}$

$$\boxed{\sigma_{\bar{x}} = 2.32}$$

7a) Given $n = 49$

Sample mean $\bar{x} = 15200$km

population mean $\bar{\mu} = 15150$ kms

Standard deviation $\sigma = 120$ km.

Test statistic $|z| = \dfrac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$$= \frac{15200 - 15150}{\frac{120}{\sqrt{49}}}$$

$$= \frac{50}{\frac{120}{7}}$$

$$= 5\cancel{0} \times \frac{7}{12\cancel{0}}$$

$$= \frac{35}{12}$$

$|z| = 2.916$

Null hypothesis H_0 : The sample is claimed from the population

Alternative hypothesis H_1 : The sample is not claimed from the population

(Two-tailed test)

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At 0.05 level critical value $\alpha = 0.05$

z value for two tailed test at 5% significance level at 0.05 is 1.96.

$z_{cal} = 2.916$

$\because z_{cal} > z_{tal}$.

Ho is rejected.

7b) Given

Sample of Englishmen $(n_1) = 6400$

mean $(x_1) = 67.85$

Standard deviation $(\sigma_1) = 2.56$

Sample of Austrians $(n_2) = 1600$

mean $(x_2) = 68.55$

Standard deviation $(\sigma_2) = 2.52$

Null Hypothesis (Ho) : There is no height difference between Englishmen & Austrians $\mu_1 = \mu_2$

Alternative Hypothesis (H_1) : There is austrians are taller than the Englishmen $\mu_1 < \mu_2$ (one-tailed)

Test statistic $|z| = \dfrac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$

$$= \frac{|67.85 - 68.55|}{\sqrt{\dfrac{(2.56)^2}{6400} + \dfrac{(2.52)^2}{1600}}} = \frac{|-0.7|}{\sqrt{0.001 + 0.004}}$$

17.14

$$= \frac{0.7}{\sqrt{0.005}} = \frac{0.7}{0.07} = 10$$

at $\alpha = 0.01$

z value at 0.01 of one-tailed test is 2.33.

$\because z_{cal} > z_{tal}$

So null hypothesis is rejected.

Austrians are taller than the Englishmen.

10a) Given

Height of father x	Height of son y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
65	67	-3.25	-1.5	10.56	2.25	4.875
66	68	-2.25	-0.5	5.06	5.06	1.125
67	64	-1.25	-4.5	1.56	20.25	5.625
67	68	-1.25	-0.5	1.56	0.25	0.625
68	72	-0.25	3.5	0.06	12.25	-0.875
69	70	0.75	1.5	0.56	2.25	1.125
71	69	2.75	0.5	7.56	0.25	1.375
73	70	4.75	1.5	22.56	2.25	7.125
$\Sigma x = 546$	$\Sigma y = 548$	$\Sigma X = 0$	$\Sigma Y = 0$	$\Sigma X^2 = 49.48$	$\Sigma Y^2 = 44.81$	$\Sigma XY = 21$

$$\bar{x} = \frac{\Sigma x}{N} = \frac{546}{8} = 68.25$$

$$\bar{y} = \frac{\Sigma y}{N} = \frac{548}{8} = 68.5$$

$\because$ mean is not whole number

Karlpearson coefficient of correlation

$$r = \frac{\Sigma XY - \frac{\Sigma X \Sigma Y}{n}}{\sqrt{\Sigma X^2 - \frac{(\Sigma X)^2}{n}}\sqrt{\Sigma Y^2 - \frac{(\Sigma Y)^2}{n}}}$$

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$$= \frac{21-0}{\sqrt{49.48-0}\sqrt{44.81-0}}$$

$$= \frac{21}{(7.034)(6.694)}$$

$$= \frac{21}{47.08}$$

$$\bar{r} = 0.446_{(ap)}$$

10b) Given $3x + 12y = 19$ — ①

$9x + 3y = 46.$ — ②

i) Value of correlation coefficient:

from ① x on y b_{xy}

$\Rightarrow 3x = 19 - 12y$

$$x = \frac{-12}{3}y + \frac{19}{3}$$

$$x = \frac{19}{3} - 4y.$$

$b_{xy} = -4.$

from ② y on x b_{yx}

$$3y = 46 - 9x$$

$$y = \frac{46}{3} - \frac{9}{3}x$$

$$y = \frac{46}{3} - 3x.$$

$b_{yx} = -3.$

coefficient correlation $\bar{r} = \sqrt{b_{xy} \times b_{yx}}$

$= \sqrt{-4 \times -3}$

$= \sqrt{12}$

$= 3.464$ $\boxed{r \le 1}$

Not possible $\rightarrow \boxed{r \not> 1}$ r should be less than 1

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Now take from 1 y on x. byx

$$12y = 19 - 3x$$

$$y = \frac{19}{12} - \frac{3}{12}x$$

$$b_{yx} = -3/12$$

from 2 take x on y bxy

$$9x = 46 - 3y$$

$$x = \frac{46}{9} - \frac{3}{9}y$$

$$b_{xy} = -\frac{3}{9}$$

correlation coefficient $r = \sqrt{b_{yx} \times b_{xy}}$

$$= \sqrt{-\frac{\cancel{3}}{12} \times -\frac{\cancel{3}}{\cancel{9}3}}$$

$$= \sqrt{-\frac{1}{12} \times -1}$$

$$= \sqrt{0.08}$$

$$r = 0.28$$

ii) means values of x, y

$(\bar{x}, \bar{y})$ lies on lines 1 & 2 solve 1 & 2

mean $\bar{x} = 5$

mean $\bar{y} = 0.334$

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$\Rightarrow x = \frac{46}{9} - \frac{3y}{9}$

Here, $b_{xy} = \frac{-3}{9}$

Now,

$r = \pm\sqrt{b_{xy} \times b_{yx}}$

$= \pm\sqrt{\frac{-3}{12}\left(\frac{-3}{9}\right)}$

$= \sqrt{\frac{9}{108}}$

Here r is positive.

Now, The mean values of x and y are.

$3x + 12y = 19$

$\cancel{3x}\ 12y = 19 - 3x$

$y = \frac{19-3x}{12}$

From ② $\Rightarrow$

$3\left(\frac{19-3x}{12}\right) + 9x = 46$

$\frac{19-3x}{4} + 9x = 46$

$\frac{19-3x+36x}{4} = 46$

$\cancel{19-3x}\ 19 + 33x = 184$

$33x = 184 - 19$

$33x = 165$

$x = \frac{165}{33} = 5$

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From, $y = \frac{19 - 3(5)}{12}$

$\Rightarrow y = \frac{19 - 15}{12}$

$\Rightarrow y = \frac{4}{12}$

$\Rightarrow y = \frac{1}{3}$

Hence,

r is positive, $x = 5$, $y = \frac{1}{3}$.

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SEMI-LOG PAPER (5 CYCLES X 1/10")